

# Option Trading

## Euan Sinclair

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# Also...

- The types of trading techniques that are legal differ enormously from country to country and market to market.
- What is “smart market making” in one place might be market manipulation in another.
- Always check with your compliance officer before doing anything that could be interpreted as manipulation.

# Major Takeaways

- Option trading is about trading, not options.
- Trading is about observation, not models or methods.
- Simple and robust ideas are always best.
- There are no rules but there are general principles.
- Trades won't have predictable results, but all choices have predictable consequences.

# Option Trading

## Session One: General Principles of Trading

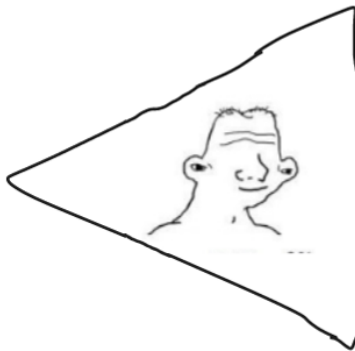
# Session One Overview

- General option trading philosophy.
- Trading Principles: edge and risk.
- The Efficient Market Hypothesis (EMH).
- Finding edge.
- Adverse selection and trading costs.

# Two (Opposite) Mistakes

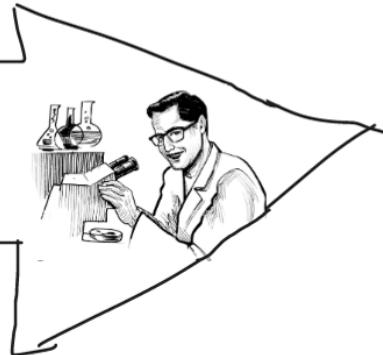
- Too naïve: it is easy to come up with ways to make money.
- Too pessimistic: it is nearly impossible to make money.

Moron



Thinks there's edge everywhere  
Fails because he tries to harness "edges"  
that don't even exist.

Theorist



Treats the market as a machine he can figure out  
Fails because he doesn't understand trading is a pragmatic  
discipline and the market is an adaptive human system.

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*We want to be here.*

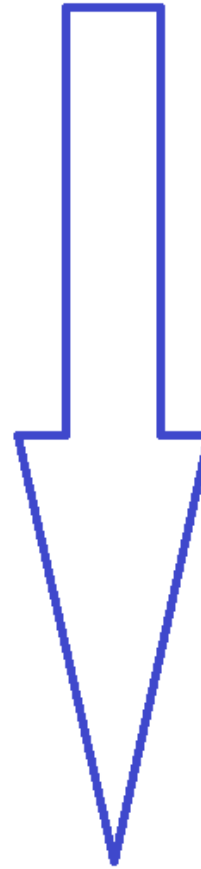
*Understand the market is a competitive adaptive  
system and it's hard to get persistent edge.*

*We want to orient our trading around obvious,  
observable, indisputable observations.*

# What is Important?

- Knowledge
- Technique
- Implementation

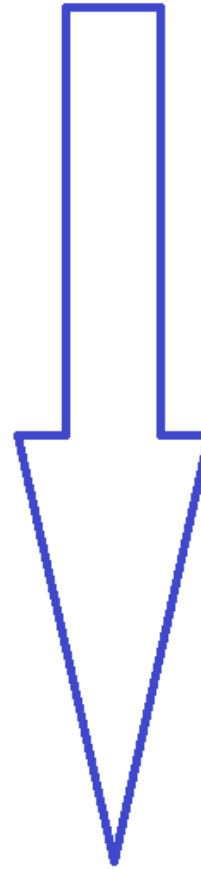
• Retail



# What is Important?

- Implementation
- Technique
- Knowledge

- Institutional



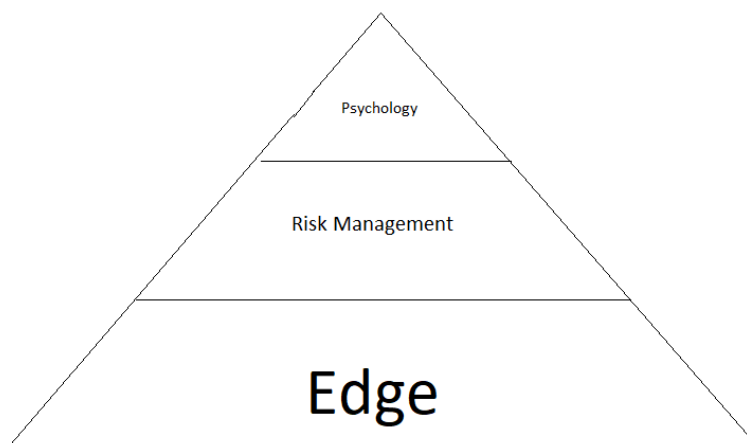
# Philosophy on Trading

- You are a trader first, an option trader second. Sometimes options will not be the best way to monetize an idea.
- Remember the goal is to make money. People can sometimes become too focused on one idea, method or problem, particularly if it is interesting to them.

# Philosophy on Math

- Quantitative traders can become too in love with the math.
- Math doesn't matter intrinsically; it is just a means to an end.
- All good solutions should be (somewhat) intuitive and robust with respect to the math used to formulate the problem.
- Math (and ML/AI) won't directly find ways to make money.
- Never start with math. Always let the phenomena lead.

# The Trading Pyramid



- Make money: done though identifying trades where we have an advantage, edge.
- Keep money: done by managing risks.
- Keep going: done by working hard and consistently.

# What is a Good Trade?

- A good trade is one that we would repeat no matter the result.
- Requires positive expectation AND an acceptable level of risk.
- EV is risk-agnostic but “acceptable risk” varies between traders and firms.

# Edge

- Edge refers to trades with positive expected value.
- Expected value (EV) is the sum of probability weighted outcomes.

$$EV = \sum_i p_i W_i$$

# Edge Example

- We have a coin that comes up heads 55% of the time.
- If we win (by calling heads) we get a dollar, and if we lose, we forfeit a dollar.

$$EV = 0.55 \times \$1 - 0.45 \times \$1$$
$$EV = \$0.1$$

- We have positive EV, so this is a potentially attractive trade.
- Note that EV is here not a result that any one trial can have.

# Edge Examples

- Bet \$1 on this coin 1000 times.
- Our average profit will be \$100.
- If we have another coin that only comes up heads 10% of the time but we win \$10 when it does, we still have  $EV = \$0.1$ .
- These bets are obviously very different. EV doesn't tell you **all** you need.

# Edge

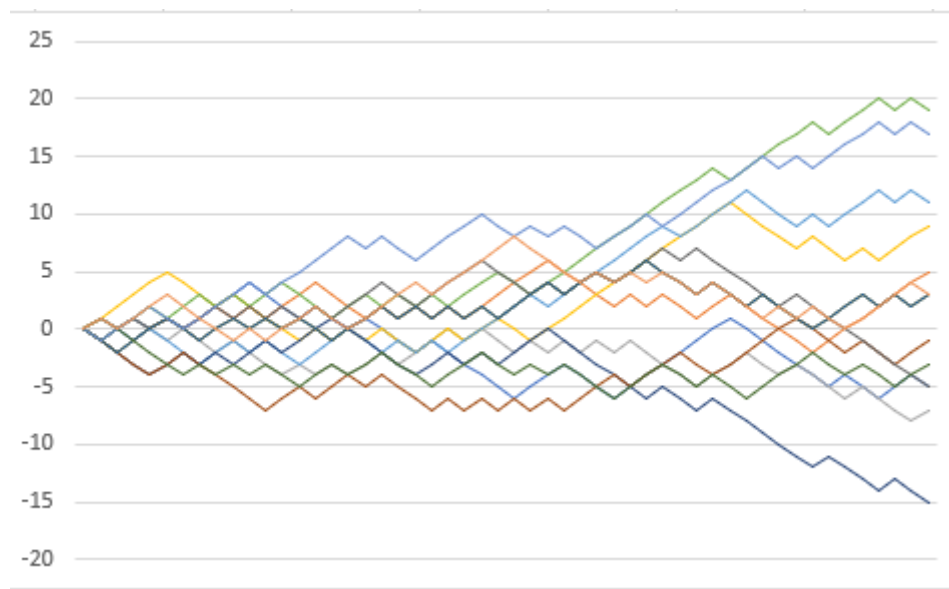
- Having an edge does not necessarily mean you win more *often* than you lose. It means you win more than you lose on *average*.
- Consider a (six-sided) die. Outcomes of one through four occur 2/3rds of the time. But knowing this is the most likely occurrence doesn't give you an edge.
- If you only get 40c for landing on 1,2,3 or 4 and lose \$1 if you don't, then you have negative edge.
- People often confuse winning percentage with edge. Betting on favorites is often a long-term loser.

# Risk

- **Risk is everything that isn't edge.**
- Model risk: we are just wrong.
- Variability of results (luck).
- Stuff we aren't trying to predict, but others might be.
- Meta-risks.
- Black swans.

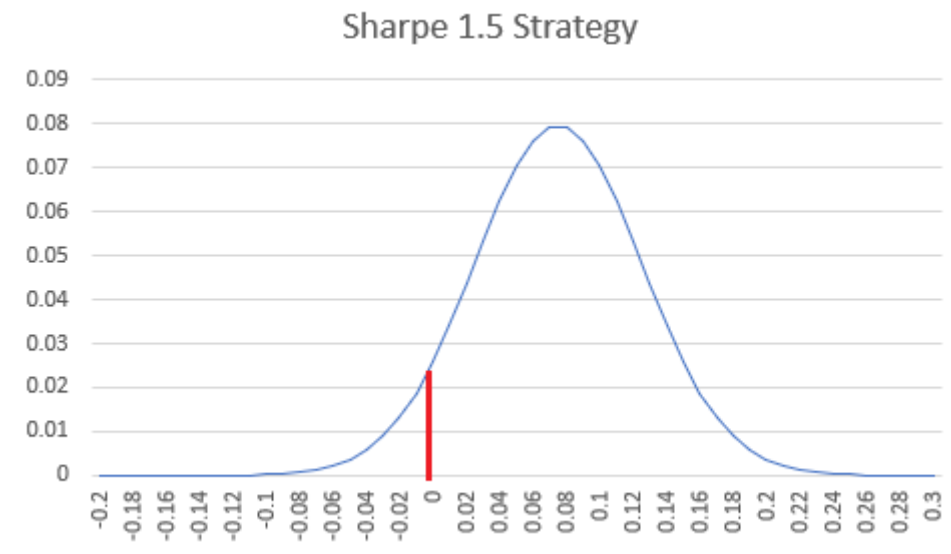
# Variance or Luck

- Even when we have edge, we can have bad results.
- Example: 12 instances of a 52% winning strategy over 52 trades.



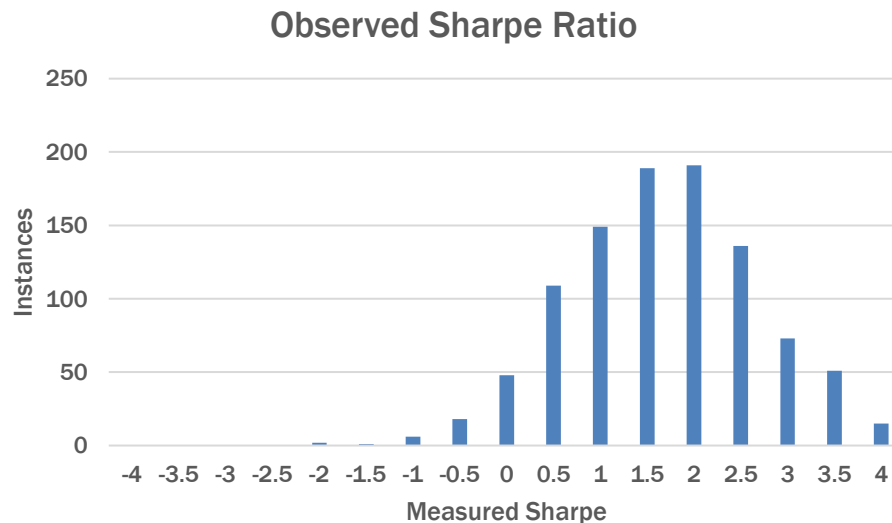
# Variance or Luck

- A true Sharpe ratio of 1.5 is good.
- But this means 7% of the time you will be a loser after a year.



# Variance or Luck

- 17% of the time you will have a Sharpe ratio less than 0.5.



# Stuff we Don't Predict

- No one can predict all the things that drive a market.
- We will be trying to predict volatility, but options also depend on currency, interest rates, dividends and stock movement.
- Try to hedge as much as this as we can.

“Isolate the edge”

# Meta-Risks

- Unlike sports or casinos, we are exposed to stuff outside the direct confines of our market.
- Bankruptcies.
- Disruptive technologies.
- Inflation.
- Fraud.
- Global economic events.
- Wars.

# Black Swans

- Events so unpredictable that they are beyond imagining.
- No one would have asked “what is the chance of there being a black swan?”, because the question would never occur to anyone.
- You cannot know in advance what a black swan will be (you can’t envisage things you can’t imagine).
- Mine have included:
  - ETF restructuring.
  - Index restructuring.
  - Covid.

# Efficient Market Hypothesis

- Formally, a market is efficient with respect to some information, if excess profits can't be derived from that information.
- Informally it means making money is hard.
- EMH is a very good approximation, but it isn't totally true. It is neither impossible nor easy to make money.

# Efficient Market Hypothesis



Pre-EMH



EMH



Reality

# Efficient Market Hypothesis

- Formally, a market is efficient with respect to some information, if excess profits can't be derived from that information.
- Informally it means making money is hard.
- EMH is a very good approximation, but it isn't totally true. It is neither impossible nor easy to make money.

"WELL ACTUALLY..."



(Don't be this guy.)

# No-Arbitrage Pricing

- An arbitrage is a risk-free profit. EMH says they don't exist (In real life they are rare.).
- Sometimes useful to assume we have found one and see where that leads us.
- This is the basis of futures pricing, option pricing and many other ideas.

# Exceptions to EMH

- Exceptions are either inefficiencies or risk premia.
- Exceptions require some of:
  1. A market constrained by costs, barriers to entry or capital constraints.
  2. Information asymmetries.
  3. Participants with different goals (remember an edge to you could be someone else's risk).

# The Categories of Edge

## Inefficiencies

- Wrinkles or inconsistencies.
- Examples:
- Products that can be replicated at better prices. A convertible bond is a bond and an option. An inefficiency would arise if the bond and option could be bought more cheaply than the convertible.
- On the run bonds are more desirable than off the run bonds and trade richer even though they have the same risks.
- Inefficiencies are often transient and not scalable.

# The Categories of Edge

## Risk Premia

- People pay to avoid certain risks. For example, we buy house insurance.

## Trading Examples:

- Equity premium: stocks go up.
- Volatility premium: options are over-priced.
- Skew premium: puts are over-priced relative to calls.
- Risk premia are time varying, but persistent. They can form the foundation of a trading business.
- Easy to find but hard to manage.

# NOT Edges

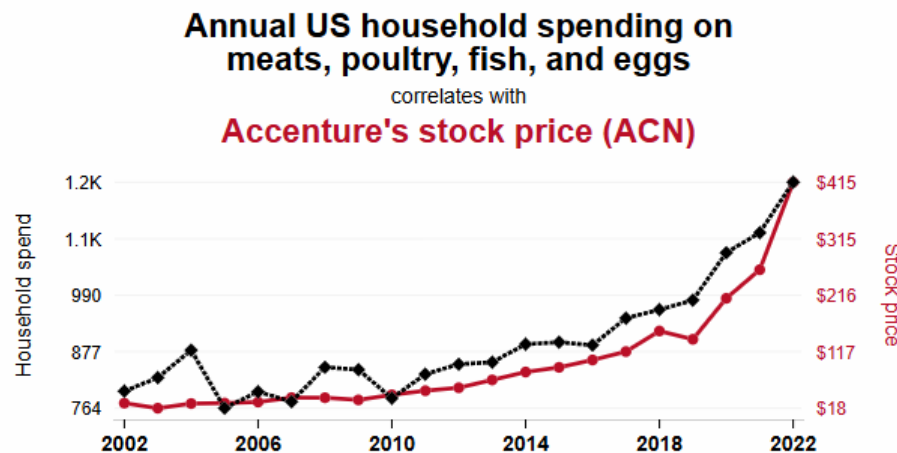
- “Feel” or intuition. You need to be able to test and quantify.
- Classic technical analysis: trend lines, chart patterns etc.
- Casual analysis of public information (newspapers etc).
- Risk management.
- Mathematical techniques or ML.

# Fundamentals are Hard

- The study “When a Crystal Ball Isn’t Enough to Make You Rich” by Victor Hagani and James White, gave participants *tomorrows* headlines.
- Trading on these earned a statistically insignificant amount and half the people *lost* money.
- ([https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=4965616](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4965616))

# P-Hacking

- Try enough things and some will work, even out of sample.



# P-Hacking

- Good science means you start with a well-formulated hypothesis.
- Particularly with volatility, nothing is really “out of sample” so statistics can’t be only arbiter.

# P-Hacking

- Your feelings about a data-driven strategy will only depend on PL, which is random in short-term.
- You will learn a lot more if you observe, postulate then test.
- You will learn what other places the idea can be applied.
- If statistics “prove” something, it will have become well known and start to decay.

# Backtesting

- Backtesting usually has a poor ROI.
- Testing option strategies has technical challenges, particularly cleaning data, huge data sets, interpolating between different options and calculating derived variables.
- More prone to data mining than delta-one products.
- Nothing is really out of sample; you can't go back to a state of zero knowledge.
- Options also depend on counter-factual history, the things that didn't happen.
- Life is too short, so test to confirm, not to explore.

# Think Like a Businessperson

- If your idea can't be reduced to a compelling elevator pitch, it is probably not worth pursuing. "I supply liquidity by looking to buy extreme dips and sell extreme spikes" summarizes an idea that might work.
- Would you invest in your idea if someone else proposed it?
- Is your idea a service to the world, doing something hard that counterparties want?

GOOD



BAD



# Finding Trades with Edge: Books

- The edges found in books will be evergreen ones: risk-premia and various forms of carry.
- Even if not directly actionable, it is vital to know the tides you might be swimming with or against.
- “Expected Returns” by Antti Ilmanen.
- “The Handbook of Equity Market Anomalies” by Leonard Zacks.
- “Positional Option Trading” by me.

# Finding Trades with Edge

- Being taught by another trader (the method used by banks and trading firms).
- Twitter follows.
- 3<sup>rd</sup> party courses (like this one).
- Reading academic papers: [www.ssrn.com](http://www.ssrn.com), arxiv.org, scholar.google.com.
- Published papers won't tell you entire strategies. But they will give you ideas and parts of strategies.

# Inefficiency Hunting: ARCTIC

- Anomaly.
- Rationale.
- Counterparty.
- Threats.
- Incentives.
- Considerations.

# Anomaly

- By definition, an inefficiency is unusual.
- To spot unusual, you need a very good appreciation of what is usual.
- For example, you only know if stocks jumping in the last few minutes of the month is “weird” if you know what happens at the close other days.

# Rationale

- “Why might this work?”
- “How might this work?”
- “Is there actually a reason why this situation makes sense?”
- Unusual behaviour on the last Friday of a month? Maybe...
- Unusual behaviour on the second Tuesday of every second month? Probably not.

# Counterparty

- “Who is on the other side?”
- “What might they know?”
- “Are they forced to trade (forced liquidation, regulatory constraints, window dressing, need to meet redemptions or adjust for inflows), or is it a choice?”

# Threats

- “What can go wrong?”
- “Is this seasonal or regime dependent in some way?”
- “How fast can others get to it?”
- “Is it robust to implementation issues?”

# Incentives

- “Is the counterparty actually winning as well?”
- “Are they laying off a trade?”
- “Is it window dressing?”

# Considerations

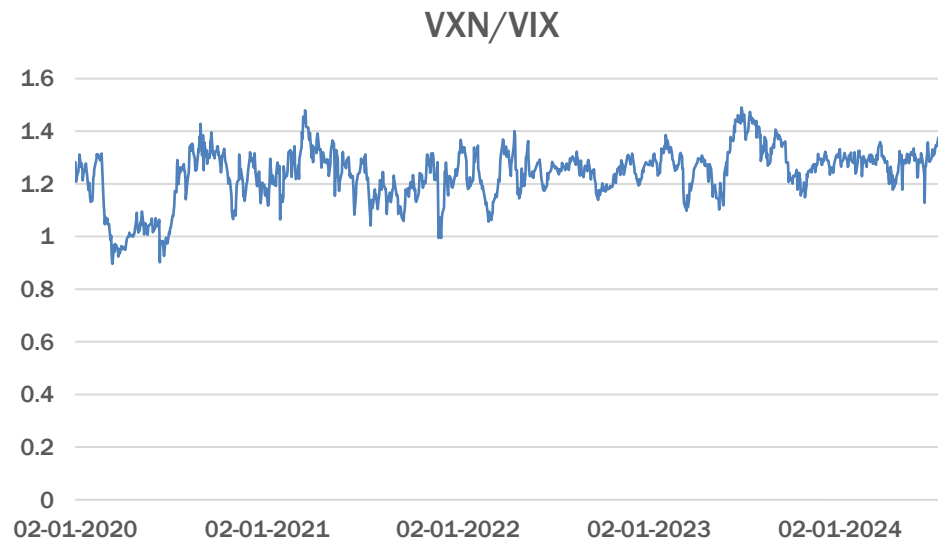
- Think about how all these pieces work together.
- (It gets easier...)

# Inefficiency Hunting: Example

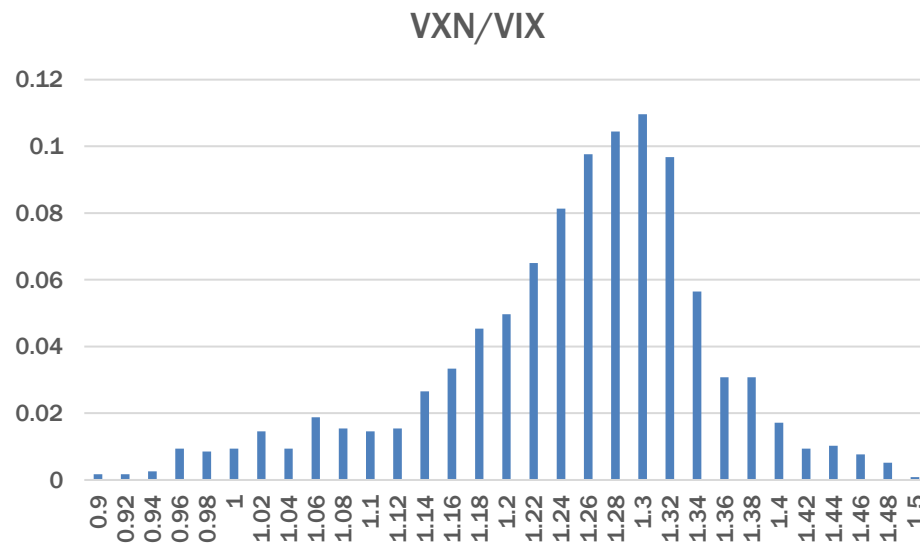
- Monday, August 5<sup>th</sup> 2024.
- Usually (almost always) NASDAQ volatility > S&P volatility.

| Statistic (VXN/VIX) | Since 2020 |
|---------------------|------------|
| Average             | 1.24       |
| Median              | 1.26       |
| Maximum             | 1.49       |
| Minimum             | 0.90       |
| Less than One       | 3%         |

# Inefficiency Hunting: Example



# Inefficiency Hunting: Example



# Inefficiency Hunting: Example

- Ratio closed on Monday at 0.91.
- Opened at 0.96 (still below 2<sup>nd</sup> percentile).
- But can't trade VIX (or VXN futures) so:
- Buy 6 QQQ 435, September 20<sup>th</sup> straddles for 36.05
- Sell 7 SPY 520, September 20<sup>th</sup> straddle for 32.36
- Exit on the 13<sup>th</sup> for profit of \$1,268.

# Inefficiency Hunting: Example

- We knew what “normal” was.
- We were plugged in enough to see the news.
- Did a very rough check of the statistics.
- Found easiest way to monetize.

# The Automation Trap

- Automate because of needs not wants.
- Automating means less active interaction.
- You learn from interaction.

# Deciding to Use a Strategy

- You want a rule like “If Sharpe  $> 2$ ” or “If drawdown is less than 15%”.
- Life is messier than that.
- There is no definitive answer, and everyone's conclusion will be different.
- It is all about trade offs.

# Plausible Reason

- Absolutely the most important consideration.
- If PL is your only feedback, you are just tracking randomness.
- “Edge” is a concept, not just a statistic.

# Confidence in Persistence

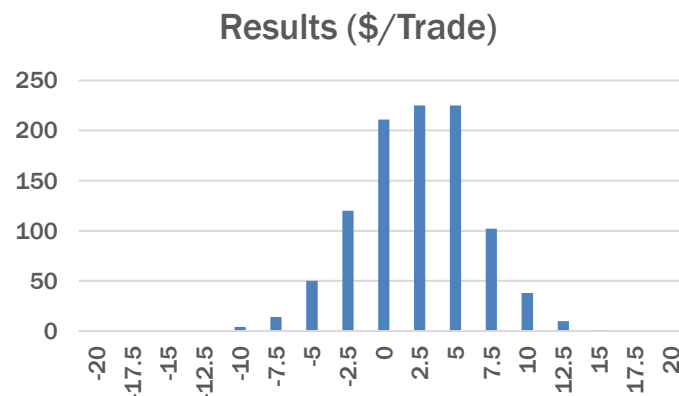
- This follows from understanding the reason.
- Risk premia are persistent.
- Inefficiencies will last until they get traded away. Is there a barrier to entry?
- (Goes back to the different ways you should trade these)

# Execution Issues

- Can you trade and clear the products needed?
- Are they liquid?
- Are they cheap to trade?
- Will adverse selection be an issue?
- Does the strategy need a lot of “hand holding”

# Reviewing Trades

- Let's say we have some historical results or a back-test (with all the caveats attached to that).
- 1000 observations:



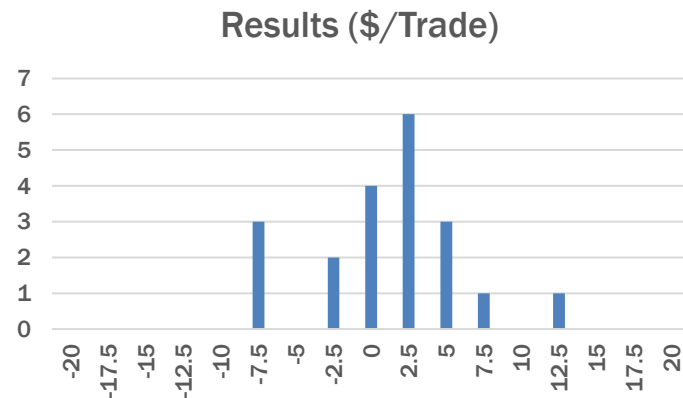
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# Reviewing Trades

- So, in the *past* (real or simulated) we had an edge.
- Now we do 20 trades.
- “I’ve learned to enter 30 minutes after the open”.
- “I’ve learned to cut losses”
- “I’ve learned to avoid Thursdays”

# Reviewing Trades

- But we are attaching stories to noise.
- 20 observations:



# Reviewing Trades

- Trading is *hard*.
- Everything is non-stationary, and your edge is small, so you need a lot of observations.
- Using results to check and improve edge is very tough.
- But you can improve other things:
  - Execution.
  - Intraday variance and how you react to it.
  - Adverse selection.